TinyBolus Formulary

Peer-review document

Version 0.0.56 · Updated 2026-07-01

Published by Aether Somnus

Developed by Sérgio Pinto, M.D. and Patrícia Lima, M.D.

Contact: tinybolusfeedback@gmail.com
Source: github.com/SGP-Prime/tinybolusformulas
Privacy policy: sgp-prime.github.io/tinybolus-site/privacy.html

---

This is a reference tool, not a prescribing system. Verify every value against your hospital protocols, product labelling, and the individual patient in front of you before administration.

About this document

This is the human-readable export of the TinyBolus paediatric reference formulary, organised section-by-section as it appears in the application. Each item is presented with its dose specification, applicable age brackets (where present), route, clinical notes, and numerical references pointing into the bibliography at the end of the document.

How to read each item

Items fall into a small number of shapes that map directly onto how the application computes them:

- **Single dose** (e.g., 10 µg/kg): one factor multiplied by weight, often with a hard maximum cap.
- **Dose range** (e.g., 2–3 mg/kg): a low–high range, also weight-scaled, optionally with a maximum cap on either bound.
- **Age-bracketed**: different dose factors for different age cohorts. Rendered as a small table per item.
- **Age-bracketed fixed value**: not weight-scaled, just a string per age band (e.g., laryngoscope blade size).
- **Computed**: piecewise logic that doesn't reduce to a single formula (ETT sizing, fluid maintenance, LMA / supraglottic airway sizing, vital-sign normal ranges).

Hard floors / contraindications

Some items carry an *age* or *weight* floor below which the formulary explicitly declines to suggest a dose. In the app this renders as `∅` (empty-set glyph) with a "not indicated" explanation. Items with such floors are flagged in red below each row.

References

Every dose / range / age bracket / sizing rule carries one or more citations. References are numbered in encounter order throughout this document, with a full bibliography at the end. The numbers in [brackets] after each item refer to entries in that bibliography.

Source

The formulary itself lives at github.com/SGP-Prime/tinybolusformulas as a single machine-readable JSON file. This document is regenerated from that file on demand by `tool/generate_formulary_review_html.dart`; do not hand-edit it.

Contents

EMERGENCY — PALS

Compression rate

VALUE `100–120/min`

References: [1]

Compression depth

RULE `4 cm <1 y · 5 cm ≥1 y`

not exceeding 6 cm at any age

References: [1]

C:V ratio (2 rescuers)

VALUE `15:2`

1 rescuer: 30:2

References: [1]

Defibrillation

DOSE `4 J/kg`

escalate to 8 J/kg (max 360 J) for refractory VF/pVT from the 6th shock

References: [1], [2]

Cardioversion (synchronised)

DOSE `1 J/kg`

double per attempt to max 4 J/kg

References: [1]

Adrenaline (cardiac arrest)

IV/IO

DOSE `10 µg/kg (max 1 mg)`

repeat every 3–5 min

References: [1], [3]

Amiodarone

IV

DOSE `5 mg/kg (max 300 mg)`

max 150 mg for subsequent doses

References: [1]

Atropine

IV

≤11 y 20 µg/kg (max 500 µg)

≥12 y 20 µg/kg (max 1 mg)

20 µg/kg · max single dose 0.5 mg child / 1 mg adolescent · cumulative max 2 mg · no minimum (Barrington 2011)

References: [1], [4], [5]

Adenosine (1st dose)

IV

DOSE `0.1 mg/kg (max 6 mg)`

References: [1]

Adenosine (2nd dose)

IV

DOSE `0.3 mg/kg (max 18 mg)`

if SVT persists, escalate in 0.05–0.1 mg/kg steps to a max single dose of 0.5 mg/kg (ERC 2025)

References: [1]

Vital Signs

Heart Rate

≤1 mo	120–170 bpm
2–12 mo	100–160 bpm
1–2 y	98–140 bpm
3–5 y	80–120 bpm
6–11 y	75–118 bpm
≥12 y	60–100 bpm

References: [6]

Systolic BP

≤12 mo	72–104 mmHg
1–2 y	86–106 mmHg
3–5 y	89–112 mmHg
6–9 y	97–115 mmHg
10–11 y	102–120 mmHg
12–15 y	110–131 mmHg
≥16 y	90–140 mmHg

References: [6]

Diastolic BP

≤12 mo	37–56 mmHg
1–2 y	42–63 mmHg
3–5 y	46–72 mmHg
6–9 y	57–76 mmHg
10–11 y	61–80 mmHg
12–15 y	64–83 mmHg
≥16 y	60–90 mmHg

References: [6]

Respiratory Rate

≤1 mo	40–60 /min
2–12 mo	30–53 /min
1–3 y	22–37 /min
4–5 y	20–28 /min
6–12 y	18–25 /min
≥13 y	12–20 /min

References: [6]

Tidal Volume

DOSE `5–7 mL/kg`

intraoperative ventilation, healthy lungs · use lower end (5 mL/kg) for neonates and very young infants

References: [7], [8]

Induction

Propofol

IV

≤1 mo	1–2.5 mg/kg
2–12 mo	3–4 mg/kg
1–6 y	3–3.5 mg/kg
7–12 y	2.5–3 mg/kg
≥13 y	2–2.5 mg/kg

after a sevoflurane inhalational induction, roughly halve the dose (lower requirement)

References: [9], [10], [11], [12]

Ketamine

IV

DOSE `1–2 mg/kg`

References: [13], [14]

Ketamine (IM)

IM

DOSE `4–5 mg/kg`

induction/sedation when no IV access · onset 2–5 min

References: [13], [14]

Rocuronium

IV

DOSE `0.6–1.2 mg/kg`

intubation 0.6 · RSI 1.0–1.2

References: [15], [16]

Succinylcholine (RSI)

IV

≤12 mo 2–3 mg/kg — *raised in infants — larger extracellular-fluid volume of distribution (Meakin 1989)*

≥1 y 1–2 mg/kg

intubating dose · laryngospasm + IM doses in Respiratory

References: [16], [6], [17]

Fentanyl (induction)

IV

≤1 mo 1–3 µg/kg — *neonate: reduce dose · inject slowly (chest-wall rigidity) · monitor for apnoea (SPA 2019)*

≥0 y 1–3 µg/kg

References: [14], [18]

Midazolam

IV

≤6 mo 0.1 mg/kg — *young infant <6 mo: reduced clearance — prolonged effect · slow injection · hypotension, esp. with an opioid*

≥0 y 0.1 mg/kg

CAP `max 5 mg`

References: [14]

Midazolam (oral premed)

PO

DOSE `0.5 mg/kg (max 20 mg)`

References: [14]

Midazolam (buccal premed)

BUCCAL

DOSE `0.3 mg/kg (max 10 mg)`

buccal alternative to oral premed (e.g. swallowing refused) · onset ~10–15 min

References: [14]

Midazolam (IN premed)

IN

DOSE `0.2–0.3 mg/kg (max 10 mg)`

intranasal premedication · 0.3 mg/kg for faster, more reliable onset

References: [19], [14]

Ondansetron

IV

DOSE `0.1 mg/kg (max 4 mg)`

References: [20], [14]

Dexamethasone (PONV)

IV

DOSE `0.15–0.25 mg/kg (max 4 mg)`

PONV prophylaxis at induction · 0.25 mg/kg effective ceiling (no benefit above)

References: [20], [21], [14]

Dexamethasone (anti-inflammatory)

IV

DOSE `0.5 mg/kg (max 8 mg)`

airway oedema / post-extubation stridor · 0.5 mg/kg/dose (vs the 0.15 mg/kg PONV dose above)

References: [22], [23]

Etomidate

IV

DOSE `0.2–0.3 mg/kg`

haemodynamically stable induction

References: [14], [6]

Atracurium

IV

DOSE `0.4–0.5 mg/kg`

intubating dose · 0.3–0.4 mg/kg in infants

References: [14], [6]

Dexmedetomidine (IN premed)

IN

DOSE `1–3 µg/kg (max 200 µg)`

intranasal premedication · 2 µg/kg usual, up to 3 in older/under-sedated · onset 30–45 min

References: [24], [25]

Painkillers

Paracetamol

IV

≤10 kg 7.5 mg/kg

>10 kg 15 mg/kg

CAP `max 1 g`

IV dosing splits by weight, not age (SmPC): ≤10 kg → 7.5 mg/kg

References: [26], [14], [27]

Ketorolac

IV

DOSE `0.5 mg/kg (max 30 mg)`

Contraindicated <6 months

References: [28], [14]

Morphine

IV

≤1 mo 0.03–0.05 mg/kg

≥0 y 0.1 mg/kg

References: [29], [14]

Fentanyl (analgesia)

IV

≤1 mo 0.5–1 μg/kg — *neonate: reduce dose · inject slowly · monitor for apnoea (SPA 2019)*

≥0 y 0.5–1 μg/kg

References: [14], [30], [18]

Metamizole

IV

≤11 mo	5 mg/kg — *reduced in infants < 1 y — higher exposure per dose from immature metabolism (Ziesenitz 2019)*
≥0 y	10–20 mg/kg

CAP `max 1 g`

Contraindicated <3 months

References: [31], [32]

Ibuprofen (IV)

IV

DOSE `10 mg/kg (max 400 mg)`

References: [33], [14]

Ibuprofen (oral)

PO

DOSE `5–10 mg/kg (max 400 mg)`

analgesic / antipyretic · typically 7–10 mg/kg/dose

References: [33], [14]

Emergency Drugs

Adrenaline (cardiac arrest)

IV/IO

DOSE `10 µg/kg (max 1 mg)`

repeat every 3–5 min

References: [1], [3]

Adrenaline (anaphylaxis IM)

IM

DOSE `10 µg/kg (max 500 µg)`

weight-based 10 µg/kg (shown) · autoinjector equivalents: <6 y 150 µg · 6–12 y 300 µg · >12 y 500 µg (EAACI)

References: [34]

Adrenaline (anaphylaxis IV)

IV

DOSE `1 µg/kg`

titrate to response · IM adrenaline is first-line

References: [1], [34]

Atropine

IV

≤11 y 20 µg/kg (max 500 µg)

≥12 y 20 µg/kg (max 1 mg)

20 µg/kg · max single dose 0.5 mg child / 1 mg adolescent · cumulative max 2 mg · no minimum (Barrington 2011)

References: [1], [4], [5]

Ephedrine

IV

≤6 mo 0.6–1.2 mg/kg — *young infants need ~10× the adult dose (indirect agent · immature SNS) · 1.2 mg/kg optimal (Szostek 2023)*

≥0 y 0.1–0.3 mg/kg

References: [35], [14], [6]

Phenylephrine

IV

DOSE `5–20 µg/kg (max 500 µg)`

vasopressor — hypotension (also tet spells)

References: [14], [6]

Lidocaine

IV

DOSE `1 mg/kg (max 100 mg)`

antiarrhythmic — alternative to amiodarone for shock-refractory VF/pVT

References: [1], [3]

Glucose 10%

IV/IO

DOSE `2 mL/kg`

0.2 g/kg for hypoglycaemia · recheck glucose after

References: [1], [5]

Calcium gluconate 10%

IV/IO

DOSE `0.5 mL/kg (max 30 mL)`

hyperkalaemia · hypocalcaemia · CCB toxicity · give slowly

References: [1], [5]

Sodium bicarbonate 8.4%

IV/IO

DOSE `1 mmol/kg`

1 mL/kg of 8.4% · for hyperkalaemia or TCA toxicity, not routine arrest

References: [1], [5]

Dantrolene

IV

DOSE `2.5 mg/kg`

repeat 5–10 min · max 10 mg/kg

References: [36], [37]

Fluids and Blood

Fluid Bolus

IV

DOSE `10 mL/kg`

References: [1]

Maintenance

RULE `Holliday-Segar 4-2-1: 4 mL/kg/h (first 10 kg) · 2 mL/kg/h (next 10 kg) · 1 mL/kg/h thereafter`

4-2-1 rule (Holliday-Segar)

References: [38], [39]

Total Blood Volume

≤1 mo	85 mL/kg
2 mo – 2 y	75 mL/kg
≥3 y	70 mL/kg

References: [6]

Tranexamic Acid

IV

DOSE `15 mg/kg (max 1 g)`

loading dose · then 2 mg/kg/h infusion (RCPCH)

References: [40], [41], [42]

Equipment

ETT

RULE `Cuffed: 3.5 + age/4, rounded to ±0.5`

≤4 KG `Neonatal weight table:`

≤1 kg	size 2.5
≤2 kg	size 3.0
≤4 kg	size 3.5

INFANT `age table (cuffed):`

≤6 mo	size 3.5
7–12 mo	size 4.0

for uncuffed tube, add 0.5 to size

References: [43]

ETT insertion depth

RULE `6 + weight (<1 mo) · weight/2 + 8 (<1 y) · age/2 + 12 (≥1 y)` *(cm)*

nasal: add 2–3 cm

References: [44], [45], [5]

---

iGel LMA

IGEL SIZING `weight bands (inclusive on both bounds):`

2–5 kg	size 1
5–12 kg	size 1.5
10–25 kg	size 2
25–35 kg	size 2.5
30–60 kg	size 3
50–90 kg	size 4
90–∞ kg	size 5

References: [46], [47]

AuraGain LMA

AURAGAIN SIZING `weight bands (inclusive on both bounds):`

2–5 kg	size 1
5–10 kg	size 1.5
10–20 kg	size 2
20–30 kg	size 2.5
30–50 kg	size 3
50–70 kg	size 4
70–100 kg	size 5
100–∞ kg	size 6

References: [48]

Laryngoscope blade

≤1 y	Miller 0–1
2–11 y	Macintosh 2
≥12 y	Macintosh 3

infants: straight (Miller) blade preferred to lift the floppy epiglottis directly · Macintosh 0–1 an accepted alternative

References: [49], [6]

Face mask

≤1 mo	Size 0
2–12 mo	Size 1
1–3 y	Size 2
4–8 y	Size 3
9–14 y	Size 4
≥15 y	Size 5

mask should cover nose and mouth from eyebrow line to chin cleft · anatomical fit is more important than estimated size

References: [6], [50]

Reservoir bag

≤6 mo	0.5 L
7 mo – 6 y	1 L
≥7 y	2 L

anaesthesia breathing-circuit bag · 0.5 L neonate/infant, 1 L to ~6 y (<25-30 kg), 2 L older child and adult

References: [51], [52]

Suction catheter

≤1 mo	6 Fr
2 mo – 1 y	8 Fr
2–8 y	10 Fr
≥9 y	12–14 Fr

roughly twice the ETT internal diameter (Fr)

References: [5], [6]

NG/OG tube

≤1 mo	6 Fr
2 mo – 1 y	8 Fr
2–8 y	10 Fr
≥9 y	12–14 Fr

References: [5], [6]

Respiratory

Salbutamol (inhaled)

INHALED

≤5 y 2–10 puffs

≥6 y 4–10 puffs

100 µg/puff · repeat incrementally as needed

References: [53], [54]

Salbutamol (nebulized)

NEBULIZED

≤5 y 2.5 mg

≥6 y 5 mg

repeat back-to-back or continuously per response

References: [53], [54]

Ipratropium (inhaled)

INHALED

≤5 y 4 puffs

≥6 y 4–8 puffs

20 µg/puff · repeat as needed

References: [55], [53]

Ipratropium (nebulized)

NEBULIZED

≤11 y 250 µg

≥12 y 500 µg

References: [14]

Magnesium Sulphate

IV

DOSE `25–50 mg/kg (max 2 g)`

References: [56], [14]

Clemastine

IV

DOSE `25 µg/kg (max 2 mg)`

References: [57], [58]

Adrenaline (nebulized)

NEBULIZED

DOSE `0.5 mg/kg (max 5 mg)`

References: [59], [14]

Succinylcholine (laryngospasm)

IV

DOSE `0.1–0.5 mg/kg`

References: [60]

Succinylcholine (IM)

IM

DOSE `4 mg/kg (max 200 mg)`

for laryngospasm or RSI without IV access · 4 mg/kg (max 200 mg)

References: [60], [61]

Rocuronium (laryngospasm)

IV

DOSE `0.1–0.2 mg/kg`

when succinylcholine is unavailable or contraindicated

References: [61]

Reversal Agents

Naloxone (titrated)

IV

DOSE `0.01 mg/kg`

post-anaesthesia reversal — titrate to respiration · repeat every 2–3 min

References: [14], [1]

Naloxone (overdose)

IV/IO/IM

DOSE `0.1 mg/kg (max 2 mg)`

suspected opioid overdose with apnoea or unresponsiveness

References: [1], [14]

Flumazenil

IV

DOSE `10 µg/kg (max 200 µg)`

max bolus 200 µg · total max 1 mg

References: [14]

Sugammadex

IV

DOSE `2–4 mg/kg`

2 mg/kg moderate block · 4 mg/kg deep block

References: [62], [63]

Sugammadex (profound block)

IV

DOSE `16 mg/kg`

for CICO / immediate reversal of full intubating-dose rocuronium

References: [64], [65]

Neostigmine

IV

DOSE `0.04–0.07 mg/kg (max 5 mg)`

coadminister with atropine to block muscarinic effects · give slowly over 1–2 min after first signs of spontaneous recovery

References: [14], [6]

Atropine (with neostigmine)

IV

DOSE `20 µg/kg (max 500 µg)`

give together with or just before neostigmine to blunt bradycardia and bronchorrhoea

References: [14], [6]

Regional Anesthesia

Lidocaine (perineural)

PERINEURAL

≤5 mo maximum 3.5 mg/kg

≥0 y maximum 5 mg/kg

References: [66], [14]

Bupivacaine (perineural)

PERINEURAL

≤5 mo maximum 1.5–2 mg/kg — *infusion: 0.2 mg/kg/h*

≥0 y maximum 2–2.5 mg/kg — *infusion: 0.4 mg/kg/h*

References: [66], [67]

Ropivacaine (perineural)

PERINEURAL

≤5 mo maximum 2 mg/kg — *infusion: 0.2 mg/kg/h*

≥0 y maximum 3 mg/kg — *infusion: 0.4 mg/kg/h*

References: [68], [69], [67]

Intralipid 20% bolus

IV

DOSE `1.5 mL/kg`

max total 12 mL/kg

References: [66], [70]

Intralipid 20% infusion

IV

DOSE `0.25 mL/min/kg`

References: [66], [70]

Weight Estimation

Used when the user does not supply a measured weight. <1 month and >12 years always require user input.

1–12 mo (infant)

FORMULA `weight (kg) = (0.5 × months) + 4`

References: [5]

1–5 y (young child)

FORMULA `weight (kg) = (2 × age) + 8`

References: [5]

6–12 y (school-age)

FORMULA `weight (kg) = (3 × age) + 7`

References: [71], [5]

Bibliography

Citations numbered in order of first encounter throughout this document.

1. Djakow J, Turner NM, Skellett S, Buysse CMP, Cardona F, de Lucas N, del Castillo J, Kiviranta P, Lauridsen KG, Markel F, Martinez-Mejias A, Roggen I, Biarent D. European Resuscitation Council Guidelines 2025: Paediatric Life Support. Resuscitation 2025;215(Suppl 1):110767.
2. Acworth J et al. ILCOR 2025 systematic review on pediatric defibrillation energy. Resuscitation Plus 2025;24:100991.
3. Lasa JJ, Dhillon GS, Topjian AA, et al. Part 8: Pediatric Advanced Life Support — 2025 American Heart Association and American Academy of Pediatrics Guidelines. Circulation 2025.
4. Barrington KJ. The myth of a minimum dose for atropine. Pediatrics 2011;127(4):783–784.
5. Advanced Life Support Group. Advanced Paediatric Life Support: A Practical Approach to Emergencies, 7th edition. Chichester: Wiley-Blackwell; 2023.
6. Coté CJ, Lerman J, Anderson BJ, eds. A Practice of Anesthesia for Infants and Children, 6th edition. Philadelphia: Elsevier; 2019.
7. Spaeth J, Schumann S, Humphreys S. Understanding pediatric ventilation in the operative setting. Part II: Setting perioperative ventilation. Paediatr Anaesth 2022;32(2):247–254.
8. Kneyber MCJ, de Luca D, Calderini E, et al. Recommendations for mechanical ventilation of critically ill children from the Paediatric Mechanical Ventilation Consensus Conference (PEMVECC). Intensive Care Med 2017;43(12):1764–1780.
9. Aun CST, Short SM, Leung DH, Oh TE. Induction dose-response of propofol in unpremedicated children. Br J Anaesth 1992;68(1):64–67.
10. Westrin P. The induction dose of propofol in infants 1–6 months of age and in children 10–16 years of age. Anesthesiology 1991;74(3):455–458.
11. van Dijk H, Hendriks MP, van Eck-Smaling MM, van Wolfswinkel L, van Loon K. Age-stratified propofol dosage for pediatric procedural sedation and analgesia. Anesth Analg 2023;136(3):551–558.
12. Kim SH, Hong JY, Suk EH, Jeong SM, Park PH. Optimum bolus dose of propofol for tracheal intubation during sevoflurane induction without neuromuscular blockade in children. Anaesth Intensive Care 2011;39(5):899–903.
13. Craven R. Ketamine. Anaesthesia 2007;62(Suppl 1):48–53.
14. Paediatric Formulary Committee. BNF for Children (online). London: BMJ Group, Pharmaceutical Press, and RCPCH Publications.
15. Taivainen T, Meretoja OA, Erkola O, Rautoma P, Juvakoski M. Rocuronium in infants, children and adults during balanced anaesthesia. Paediatr Anaesth 1996;6(4):271–275.
16. Mazurek AJ, Rae B, Hann S, et al. Rocuronium versus succinylcholine: are they equally effective during rapid-sequence induction of anesthesia? Anesth Analg 1998;87(6):1259–1262.
17. Meakin G, McKiernan EP, Morris P, Baker RD. Dose-response curves for suxamethonium in neonates, infants and children. Br J Anaesth 1989;62(6):655–658.
18. Cravero JP, Agarwal R, Berde C, et al. The Society for Pediatric Anesthesia recommendations for the use of opioids in children during the perioperative period. Paediatr Anaesth 2019;29(6):547–571.

19. Baldwa NM, Padvi AV, Dave NM, Garasia MB. Atomised intranasal midazolam spray as premedication in pediatric patients: comparison between two doses of 0.2 and 0.3 mg/kg. J Anesth 2012;26(3):346–350.

20. Gan TJ, Belani KG, Bergese S, et al. Fourth Consensus Guidelines for the Management of Postoperative Nausea and Vomiting. Anesth Analg 2020;131(2):411–448.

21. Madan R, Bhatia A, Chakithandy S, Subramaniam R, Rammohan G, Deshpande S, Singh M, Kaul HL. Prophylactic dexamethasone for postoperative nausea and vomiting in pediatric strabismus surgery: a dose ranging and safety evaluation study. Anesth Analg 2005;100(6):1622–1626.

22. Anene O, Meert KL, Uy H, Simpson P, Sarnaik AP. Dexamethasone for the prevention of postextubation airway obstruction: a prospective, randomized, double-blind, placebo-controlled trial. Crit Care Med 1996;24(10):1666–1669.

23. Malhotra D, Gurcoo S, Qazi S, Gupta S. Randomized comparative efficacy of dexamethasone to prevent postextubation upper airway complications in children and adults in ICU. Indian J Anaesth 2009;53(4):442–449.

24. Yuen VM, Hui TW, Irwin MG, Yao TJ, Chan L, Wong GL, Hasan MS, Shariffuddin II. A randomised comparison of two intranasal dexmedetomidine doses for premedication in children. Anaesthesia 2012;67(11):1210–1216.

25. Jun JH, Kim KN, Kim JY, Song SM. The effects of intranasal dexmedetomidine premedication in children: a systematic review and meta-analysis. Can J Anaesth 2017;64(9):947–961.

26. Anderson BJ. Paracetamol (acetaminophen): mechanisms of action. Paediatr Anaesth 2008;18(10):915–921.

27. Paracetamol 10 mg/mL solution for infusion — Summary of Product Characteristics. electronic Medicines Compendium (emc), current edition.

28. Gupta A, Daggett C, Ludwick J, Wells W, Lewis A. Ketorolac after congenital heart surgery: does it increase the risk of significant bleeding complications? Paediatr Anaesth 2005;15(2):139–142.

29. Bouwmeester NJ, Anderson BJ, Tibboel D, Holford NH. Developmental pharmacokinetics of morphine and its metabolites in neonates, infants and young children. Br J Anaesth 2004;92(2):208–217.

30. Association of Paediatric Anaesthetists of Great Britain and Ireland. Good Practice in Postoperative and Procedural Pain Management, 2nd edition. Paediatr Anaesth 2012;22(Suppl 1):1–79.

31. Ziesenitz VC, Rodieux F, Atkinson A, et al. Dose evaluation of intravenous metamizole (dipyrone) in infants and children: a prospective population pharmacokinetic study. Eur J Clin Pharmacol 2019;75(11):1491–1502.

32. Fieler M, Eich C, Becke K, et al. Metamizole for postoperative pain therapy in 1177 children: a prospective, multicentre, observational, postauthorisation safety study. Eur J Anaesthesiol 2015;32(12):839–843.

33. Southey ER, Soares-Weiser K, Kleijnen J. Systematic review and meta-analysis of the clinical safety and tolerability of ibuprofen compared with paracetamol in paediatric pain and fever. Curr Med Res Opin 2009;25(9):2207–2222.

34. Muraro A, Roberts G, Worm M, et al. Anaphylaxis: guidelines from the European Academy of Allergy and Clinical Immunology. Allergy 2014;69(8):1026–1045.

35. Szostek AS, Saunier C, Elsensohn MH, Boucher P, Merquiol F, Gerst A, et al. Effective dose of ephedrine for treatment of hypotension after induction of general anaesthesia in neonates and infants less than 6 months of age: a multicentre randomised, controlled, open label, dose escalation trial. Br J Anaesth 2023;130(5):603–610.

36. Hopkins PM, Girard T, Dalay S, Jenkins B, Thacker A, Patteril M, McGrady E. Malignant hyperthermia 2020: Guideline from the Association of Anaesthetists. Anaesthesia 2021;76(5):655–664.

37. Glahn KPE, Ellis FR, Halsall PJ, Müller CR, Snoeck MMJ, Urwyler A, Wappler F. Recognizing and managing a malignant hyperthermia crisis: guidelines from the European Malignant Hyperthermia Group. Br J Anaesth 2010;105(4):417–420.

38. Holliday MA, Segar WE. The maintenance need for water in parenteral fluid therapy. Pediatrics 1957;19(5):823–832.

39. Feld LG, Neuspiel DR, Foster BA, et al. Clinical Practice Guideline: Maintenance Intravenous Fluids in Children. Pediatrics 2018;142(6):e20183083.

40. Royal College of Paediatrics and Child Health. Evidence statement: major trauma and the use of tranexamic acid in children. RCPCH; November 2012.

41. Faraoni D, Goobie SM. The efficacy of antifibrinolytic drugs in children undergoing noncardiac surgery: a systematic review of the literature. Anesth Analg 2014;118(3):628–636.

42. Goobie SM, Meier PM, Pereira LM, et al. Efficacy of tranexamic acid in pediatric craniosynostosis surgery: a double-blind, placebo-controlled trial. Anesthesiology 2011;114(4):862–871.

43. Duracher C, Schmautz E, Martinon C, et al. Evaluation of cuffed tracheal tube size predicted using the Khine formula in children. Paediatr Anaesth 2008;18(2):113–118.

44. Phipps LM, Thomas NJ, Gilmore RK, et al. Prospective assessment of guidelines for determining appropriate depth of endotracheal tube placement in children. Pediatr Crit Care Med 2005;6(5):519–522.

45. Lau N, Playfor SD, Rashid A, Dhanarass M. New formulae for predicting tracheal tube length. Paediatr Anaesth 2006;16(12):1238–1243.

46. Intersurgical Ltd. i-gel User Guide: Adult and Paediatric Sizes. Intersurgical, Wokingham, UK (current edition).

47. Beringer RM, Kelly F, Cook TM, et al. A cohort evaluation of the paediatric i-gel airway during anaesthesia in 120 children. Anaesthesia 2011;66(12):1121–1126.

48. Ambu A/S. Ambu AuraGain Single Use Laryngeal Mask — Instructions for Use. V7, August 2023.

49. Passi Y, Sathyamoorthy M, Lerman J, Heard C, Marino M. Comparison of the laryngoscopy views with the size 1 Miller and Macintosh laryngoscope blades lifting the epiglottis or the base of the tongue in infants and children <2 yr of age. Br J Anaesth 2014;113(5):869–874.

50. Rendell-Baker L, Soucek DH. New pediatric face masks and anaesthetic equipment. Br Med J 1962;1(5293):1690.

51. Kalra A. Pediatric Anesthesia Digital Handbook. Department of Anesthesiology, Tufts Medical Center, Boston, MA (maskinduction.com), current edition.

52. Intersurgical Ltd. Reservoir bags (paediatric 0.5 L and 1 L; adult 2 L; 3 L also available). Intersurgical, Wokingham, UK (current edition).

53. British Thoracic Society / Scottish Intercollegiate Guidelines Network. British guideline on the management of asthma (SIGN 158). Edinburgh: BTS/SIGN; 2019, updated 2024.

54. Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2024 update. https://ginasthma.org/

55. Griffiths B, Ducharme FM. Combined inhaled anticholinergics and short-acting beta2-agonists for initial treatment of acute asthma in children. Cochrane Database Syst Rev 2013;(8):CD000060.

56. Griffiths B, Kew KM. Intravenous magnesium sulfate for treating children with acute asthma in the emergency department. Cochrane Database Syst Rev 2016;4:CD011050.

57. Ring J, Beyer K, Biedermann T, et al. Guideline (S2k) on acute therapy and management of anaphylaxis: 2021 update. Allergo J Int 2021;30(1):1–25.

58. Tavegyl (clemastine fumarate) 1 mg/mL solution for injection — Summary of Product Characteristics, current edition.

59. Bjornson C, Russell K, Vandermeer B, Klassen TP, Johnson DW. Nebulized epinephrine for croup in children. Cochrane Database Syst Rev 2013;(10):CD006619.

60. Larson CP Jr. Laryngospasm — the best treatment. Anesthesiology 1998;89(5):1293–1294.

61. Gavel G, Walker RWM. Laryngospasm in anaesthesia. Contin Educ Anaesth Crit Care Pain 2014;14(2):47–51.

62. Plaud B, Meretoja O, Hofmockel R, et al. Reversal of rocuronium-induced neuromuscular blockade with sugammadex in pediatric and adult surgical patients. Anesthesiology 2009;110(2):284–294.

63. Lee S, Chung W. Sugammadex for our little ones: a brief narrative review. Anesth Pain Med (Seoul) 2024;19(4):269–279.

64. Lee C, Jahr JS, Candiotti KA, et al. Reversal of profound neuromuscular block by sugammadex administered three minutes after rocuronium. Anesthesiology 2009;110(5):1020–1024.

65. Bridion (sugammadex) prescribing information. Merck Sharp & Dohme LLC, 2025.

66. Neal JM, Barrington MJ, Fettiplace MR, et al. The third American Society of Regional Anesthesia and Pain Medicine practice advisory on local anesthetic systemic toxicity: executive summary 2017. Reg Anesth Pain Med 2018;43(2):113–123.

67. Suresh S, Ecoffey C, Bosenberg A, et al. The European Society of Regional Anaesthesia and Pain Therapy/American Society of Regional Anesthesia and Pain Medicine recommendations on local anesthetics and adjuvants dosage in pediatric regional anesthesia. Reg Anesth Pain Med 2018;43(2):211–216.

68. Lönnqvist PA, Westrin P, Larsson BA, et al. Ropivacaine pharmacokinetics after caudal block in 1–8-year-old children. Br J Anaesth 2000;85(4):506–511.

69. Bosenberg AT, Thomas J, Lopez T, et al. Plasma concentrations of ropivacaine following a single-shot caudal block of 1, 2 or 3 mg/kg in children. Acta Anaesthesiol Scand 2001;45(10):1276–1280.

70. American Society of Regional Anesthesia and Pain Medicine. Checklist for treatment of local anesthetic systemic toxicity. ASRA Pain Medicine; 2020 revision.

71. Luscombe MD, Owens BD, Burke D. Weight estimation in paediatrics: a comparison of the APLS formula and the formula 'Weight = 3(age)+7'. Emerg Med J 2011;28(7):590–593.